# CSCI 5244: Homework Set 3

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1. **[Phase Flip Codes]** Recall the phase-flip error $\begin{cases} |0\rangle \to |0\rangle \\ |1\rangle \to -|1\rangle \end{cases}$, and the states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The following $[3, 1]$ codes can be used to detect and correct the phase-flip error:

$$\begin{cases} |0\rangle \to |\bar{0}\rangle = |+++\rangle \\ |1\rangle \to |\bar{1}\rangle = |---\rangle \end{cases}$$

   (a) Draw the quantum circuit for encoding the above phase-flip codes.

   (b) Give the stabilizers that can be used to detect the phase-flip error.

   (c) Continuing on the encoding circuit from (a), draw the circuit for error detection and correction.

2. Recall Pauli X,Y,Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

   (a) Show the following anti-communication relationships:

$$\sigma_i, \sigma_j = 2\delta_{ij} I$$

   where $\delta_{ij}$ is the Kronecker delta.